Sequence processing

- Text is discrete
- Variable length

$$\boxed{\text{Sequence}} \longrightarrow \text{Tokenization} \longrightarrow \text{Embeddings} \longrightarrow \text{Model}$$

Sequence: $(w_1, \dots, w_T)$

Tokenization: $(t_1, \dots, t_L)$ $t_i \in V$

Embeddings: $(e_1, \dots, e_L)$ $e_i \in \mathbb{R}^d$

- V - vocabulary - set of tokens

Tokenization

На дворе трава, на траве дрова.

- Character-level

|Н|а| |д|в|о|р|е| |т|р|а|в|а|,| |н|а| |т|р|а|в|е| |д|р|о|в|а|.|

\+ Easy to encode new words

\+ Very small vocab

− Loses language semantics

− Very long sequences

- Word-level

|На|дворе|трава|,|на|траве|
дрова|.|

— Very large vocab

— No tokens for new words

- <UNK> – token for unknown words

\+ Preserve language structure

\+ Shorter sequences

- Subword tokenization

<u>Byte-pair encoding (BPE)</u>

1) Start with character-level tokenization
2) while $size(V) < vocab\_size$:
   - Find most frequent pair of tokens
   - Merge 2 tokens, add to vocab, re-tokenize

0) {Н, а, д, в, о, р, е, т, н, ',', '.'}

|Н|а| |д|в|о|р|е| |т|р|а|в|а|,| |н|а| |т|р|а|в|е
|д|р|о|в|а|.|

1) {Н, а, д, в, о, р, е, т, н, ',', '.', тр}

|Н|а| |д|в|о|р|е| |т р а|в|а|,| |н|а| |т р а|в|е|
|д|р|о|в а|.|

2) {Н, а, д, в, о, р, е, т, н, ',', '.', тр, тра}

|Н|а| |д|в|о|р|е| |т р а|в|а|,| |н|а| |т р а|в|е|
|д|р|о|в а|.|

Embeddings

Word2Vec

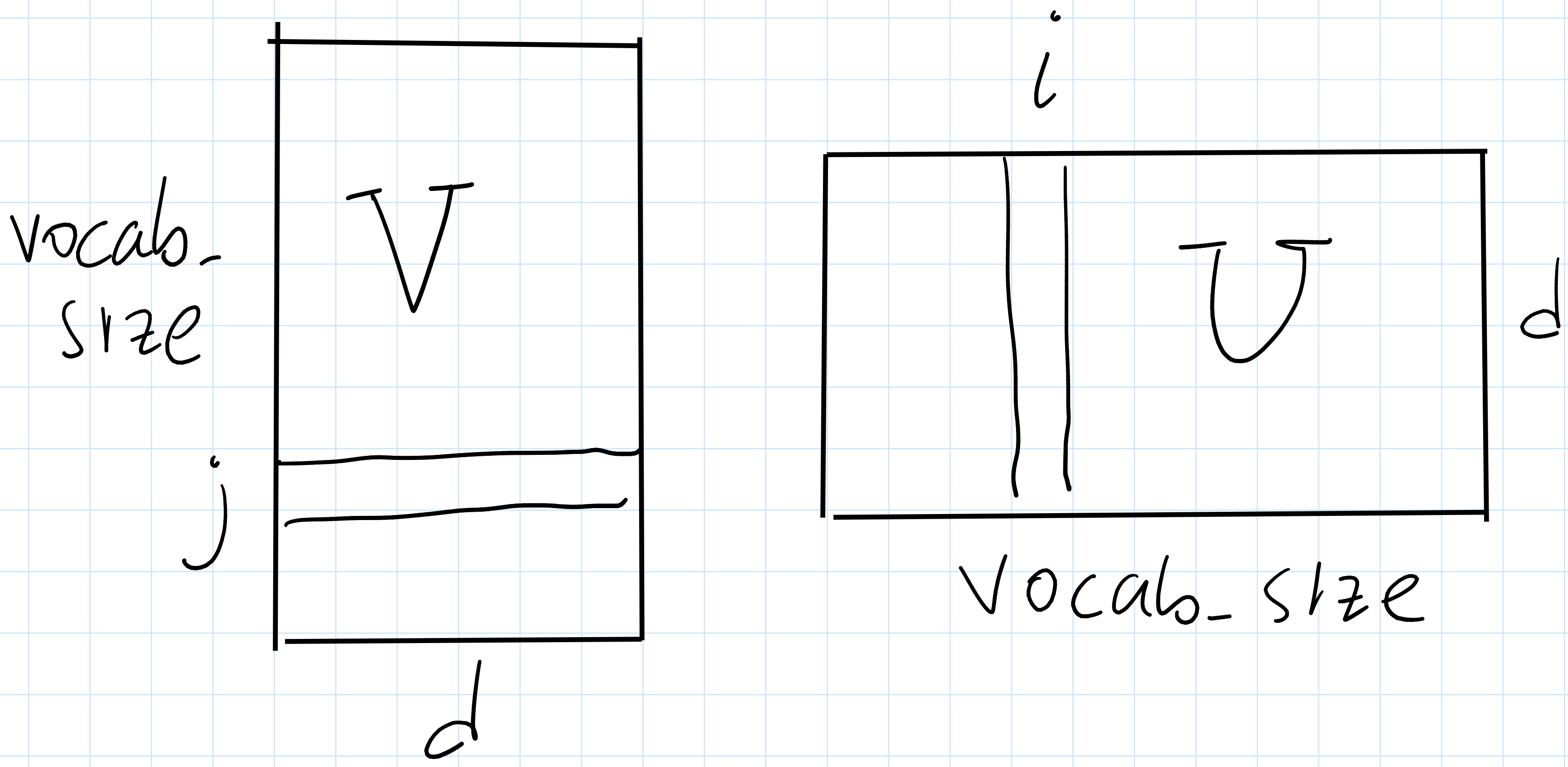
$w_1 \; [w_2 \quad w_3 \quad \boxed{w_4} \quad w_5 \quad w_6] \; w_7 \; \ldots$

- Continuous Bag of Words (CBOW)

$$p(w_4 \mid w_2, w_3, w_5, w_6)$$

- Skip-gram

$$p(w_2 \mid w_4) \quad p(w_3 \mid w_4) \quad \ldots \quad p(w_6 \mid w_4)$$



d-dim of embeds

$v_j \in \mathbb{R}^d$

$u_i \in \mathbb{R}^d$

$$p(w_i \mid w_j) = \frac{\exp(v_j^T u_i)}{\sum_{k=1}^{vocab} \exp(v_j^T u_k)}$$

$$-\sum_{w_j\text{-center}} \log p(w_i \mid w_j) \rightarrow \min_{V,U}$$ (Skip-gram)

w_j - center

w_i - context

$$p(w_j \mid w_{j-H}, w_{j-H+1}, w_{j-1}, w_{j+1}, \dots w_{j+H})$$

$$(v_{j-H} + v_{j-H+1} + \dots + v_{j+H})^T u_j$$

↑ context vector

Multiclass → Binary

w_i - context word

w_j - central word

"Do this words appear together"

$$p(w_i \text{ from context } w_j) = \sigma(v_j^T u_i)$$

$$-\sum_{\substack{w_j\text{-center} \\ w_i\text{-context}}} \log(\sigma(v_j^T u_i)) \rightarrow \min_{V,U}$$

Negative sampling

$$-\sum_{\substack{w_j - \text{center} \\ w_i - \text{context}}} \log \sigma(v_j^T u_i) - \sum_{\substack{w_i, w_j - \\ - \text{negative pair}}} \log(1 - \sigma(v_j^T u_i)) \to \min_{V, U}$$

nn.Embedding[i] : token index → embed. vector

- V
- U
- $\frac{1}{2}(V + U)$

man ↔ king

woman ↔ queen

$v_{man} - v_{king} \approx v_{woman} - v_{queen}$

Fastnet

p(дрова | трава) — skip-gram

трава → <трава>, <тр, тра, рав, ава, ва>
n-gram

p(дрова | тр)

p(=//= | тра)

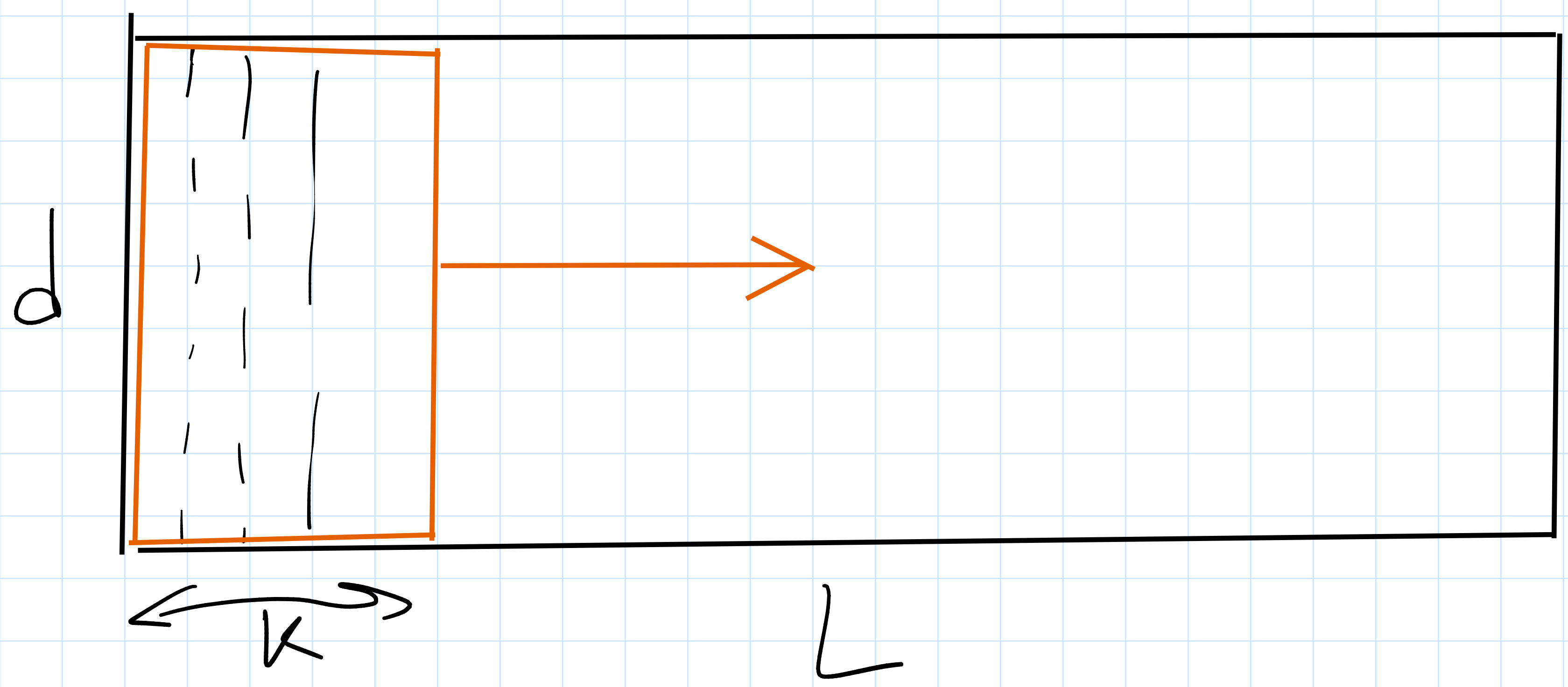
....

\+ New words

\+ Information exchange between words

Glove – Global vectors

TextCNN 1-D convolutions



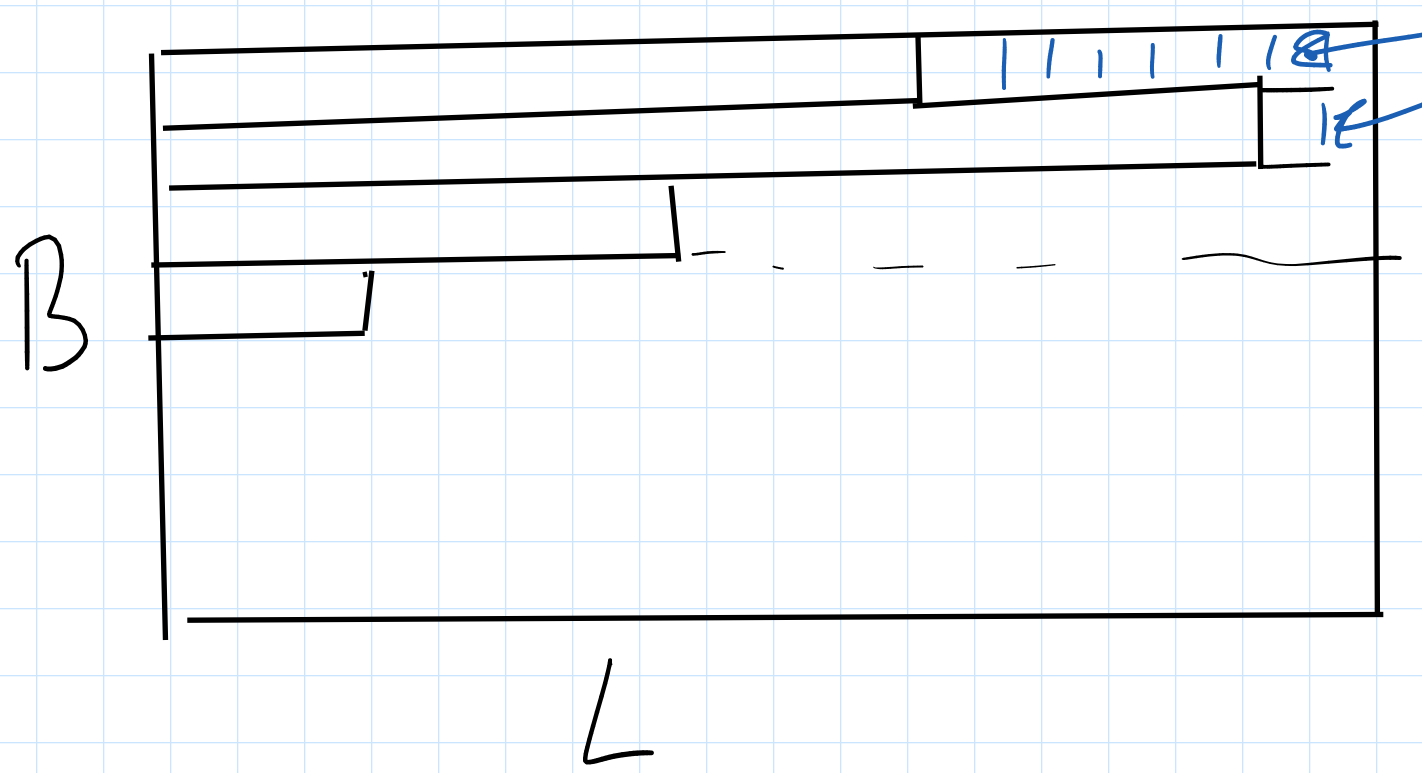
d - channels

L - spatial dimension

Kernel $d \times d' \times k \times 1$

$d \times L \to d' \times L'$

$B \times C \times H \times W$



<PAD> - padding token